



MC311 Requests Pattern Analysis

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What is
MC311?

Data Ingestion + Wrangling



- Tools:
 - Python (Google Colab)
- MC311 Requests Dataset
 - Source: datamontgomery
 - Filtered to January 2025 to April 2026
 - Filtered to 3 service areas and 5 ZIP codes
 - Created `response\_time`, `day\_of\_week`, `month`, and `season` variables



Data Ingestion + Wrangling



- **Maryland Weather Dataset**
 - Source: National Oceanic and Atmospheric Administration
 - Created `avg\_temp` variable
- **ZIP Code Tabulation Areas Dataset**
 - Source: Maryland.gov
- **Maryland Median Income Dataset**
 - Source: Data Commons

Basic Descriptive Statistics

Merged all datasets together by either date or ZIP Code

Final dataset contains:

- 10,000+ service requests
- Service areas and their respective detailed topics
- Response time and SLA variables
- Demographic and income statistics across 5 ZIP codes
- Max, min, and average temperature every day



Goals

-01-

Explore how
weather,
location, and
demographics
may influence
service
patterns

-02-

Test if factors
in dataset can
predict
service
outcomes
/efficiency

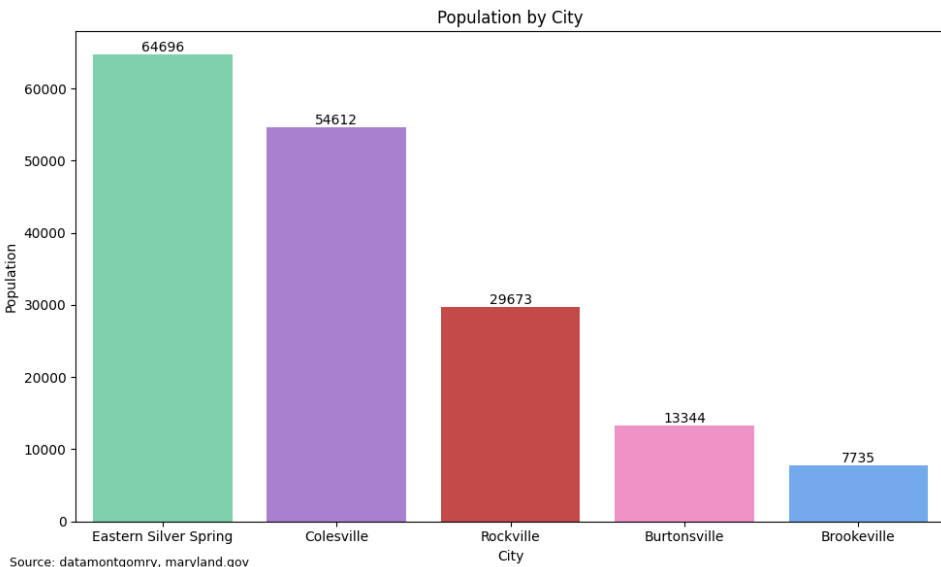
-03-

Identify any
other trends

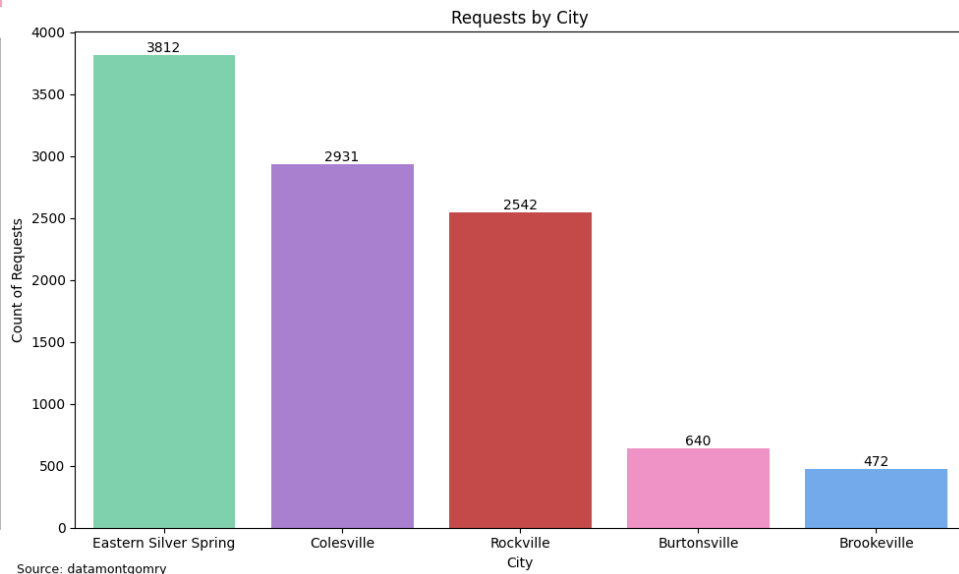


Final Data
Product

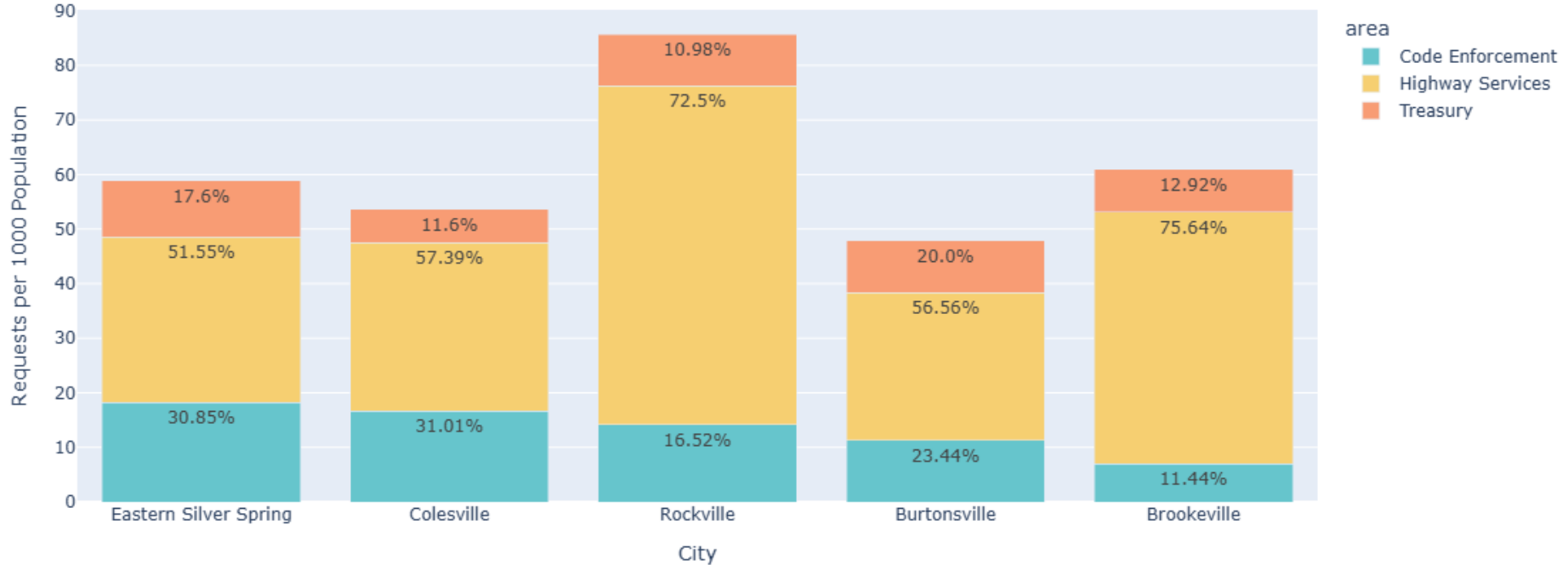
Population by City



of Requests by City



Requests per 1000 Population by City, Colored by Service Area

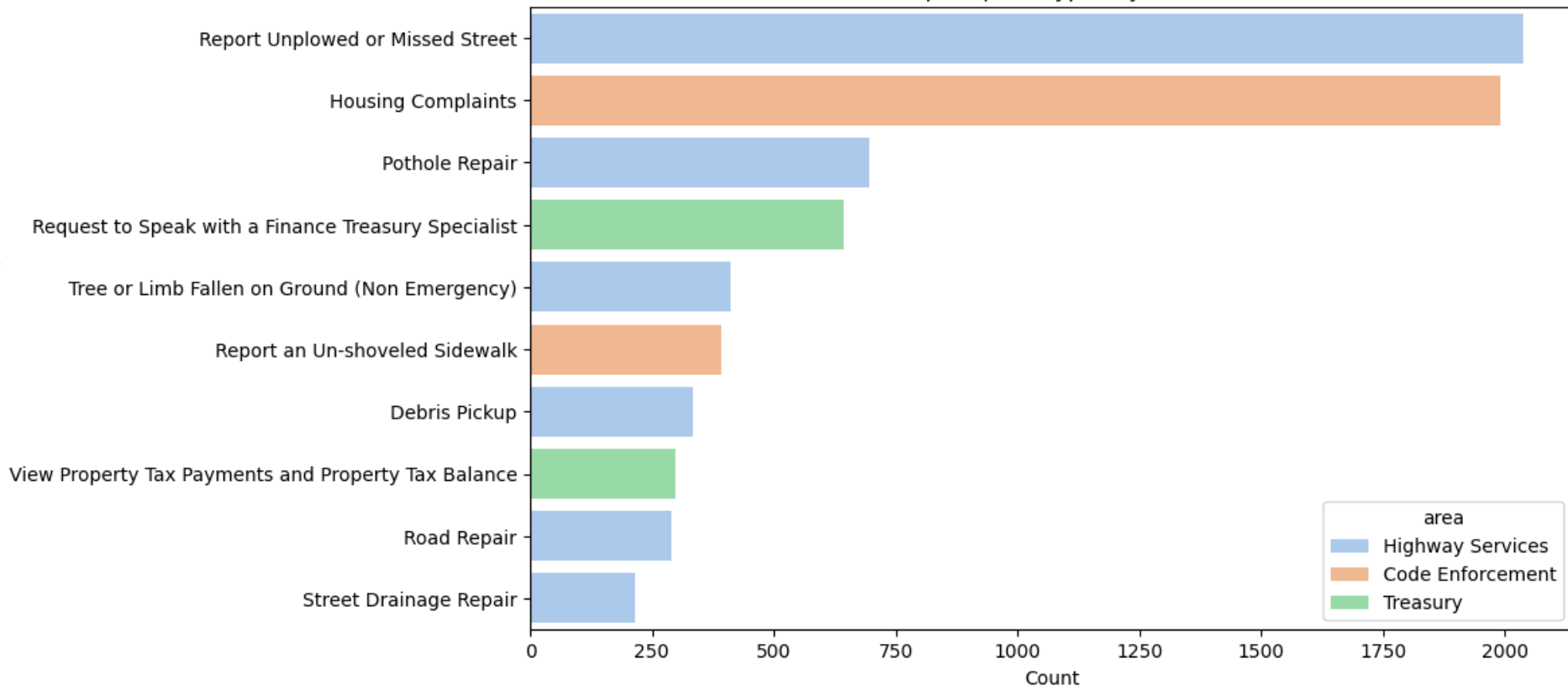


Source: datamontgomry, maryland.gov

Requests per Capita (1000 people)

- Rockville has disproportionately high request rates
- Dominance of Highway Services across cities

Top Request Types by Area

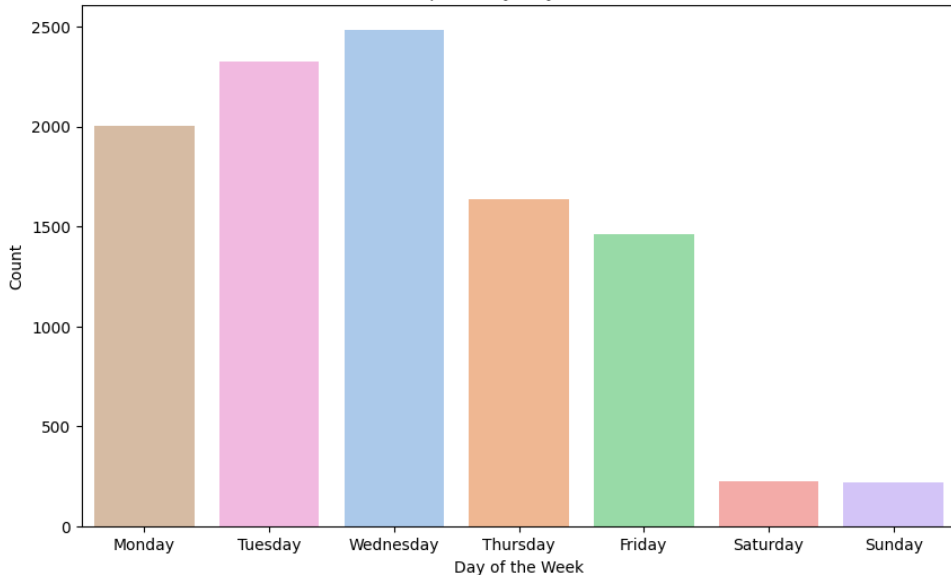


Source: datamontgomery

Top Requests Topics, Colored by Area

Requests by Day of the Week

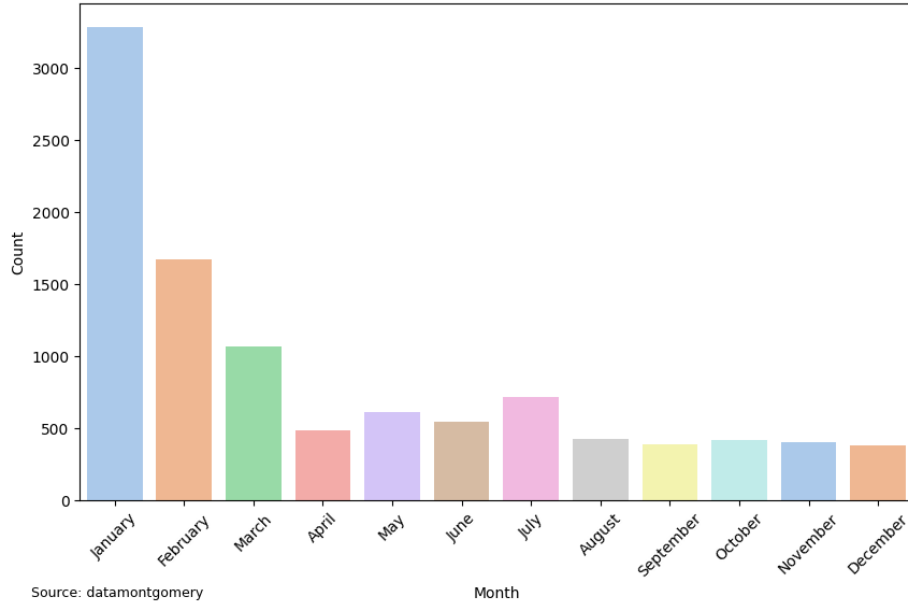
Requests by Day of Week



Source: datamontgomery

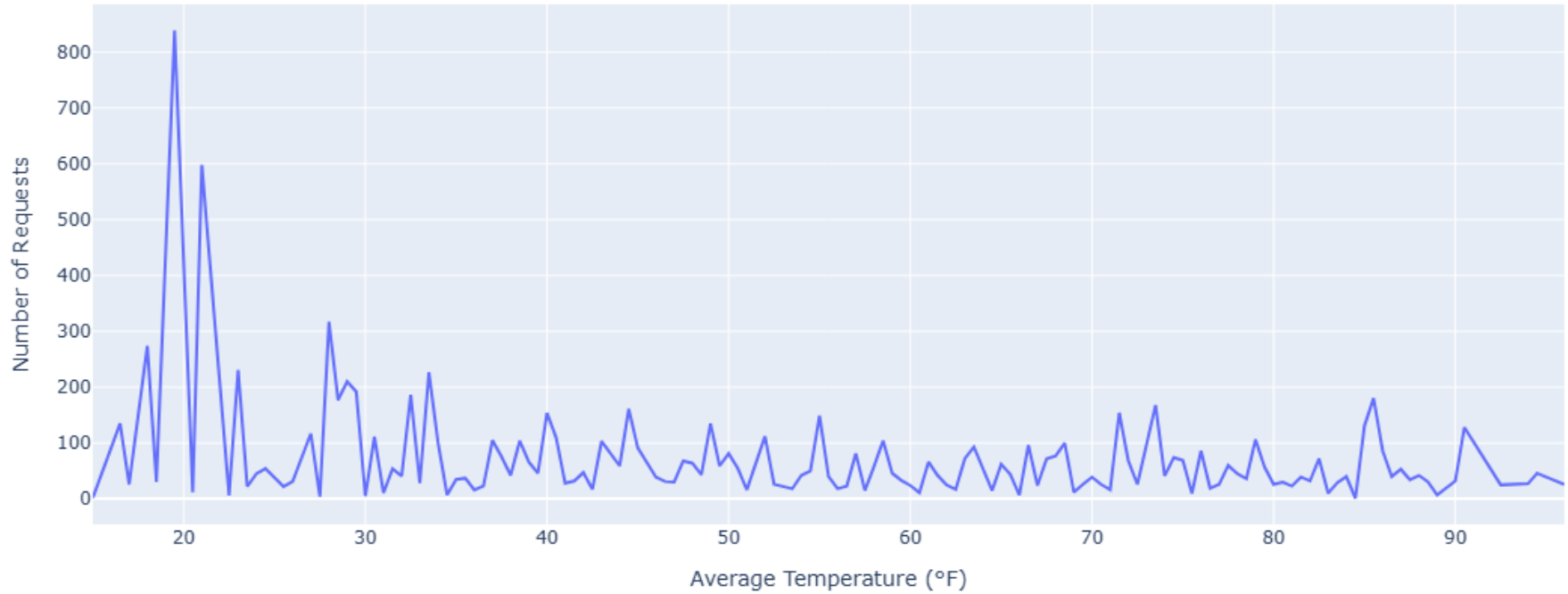
Requests by Month

Requests by Month



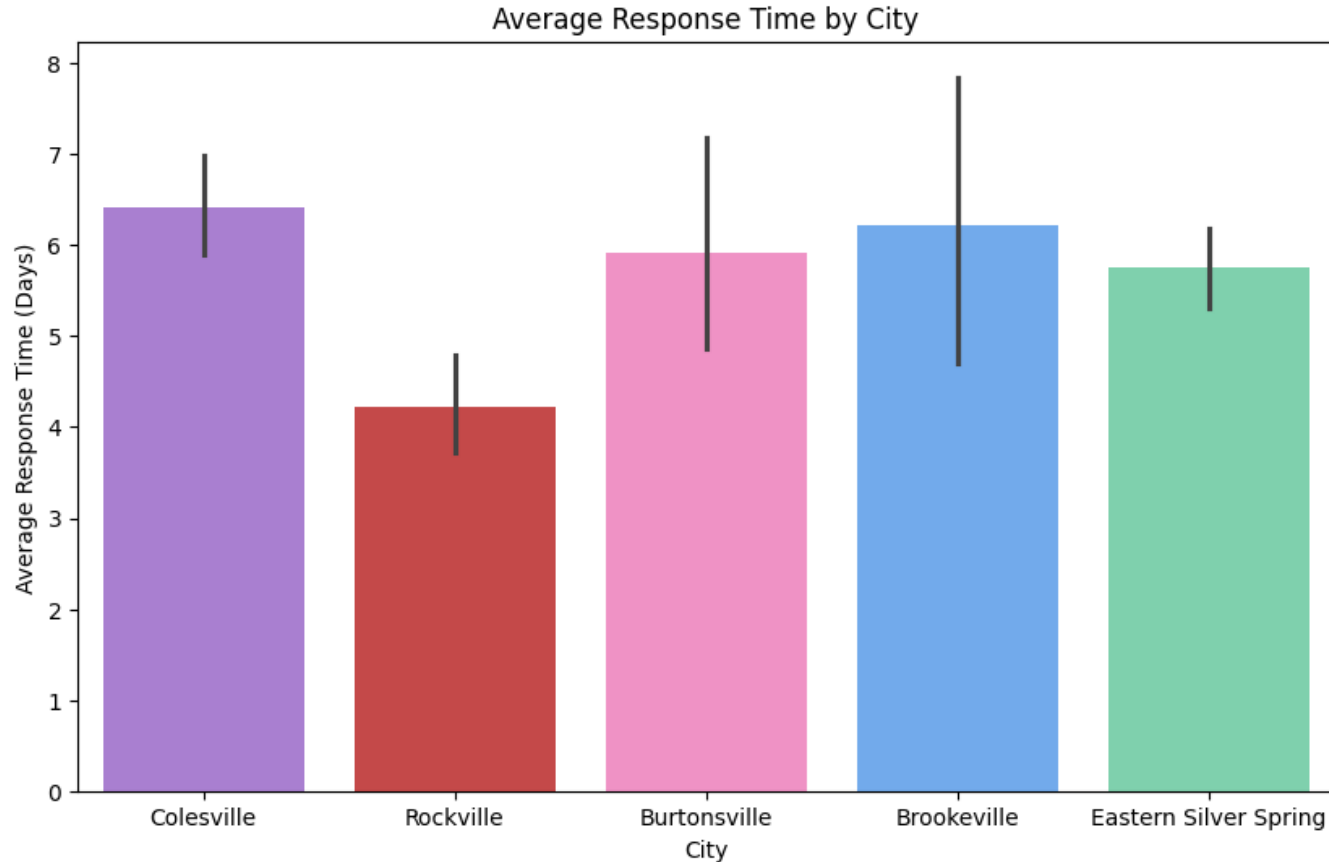
Source: datamontgomery

Requests vs Temperature



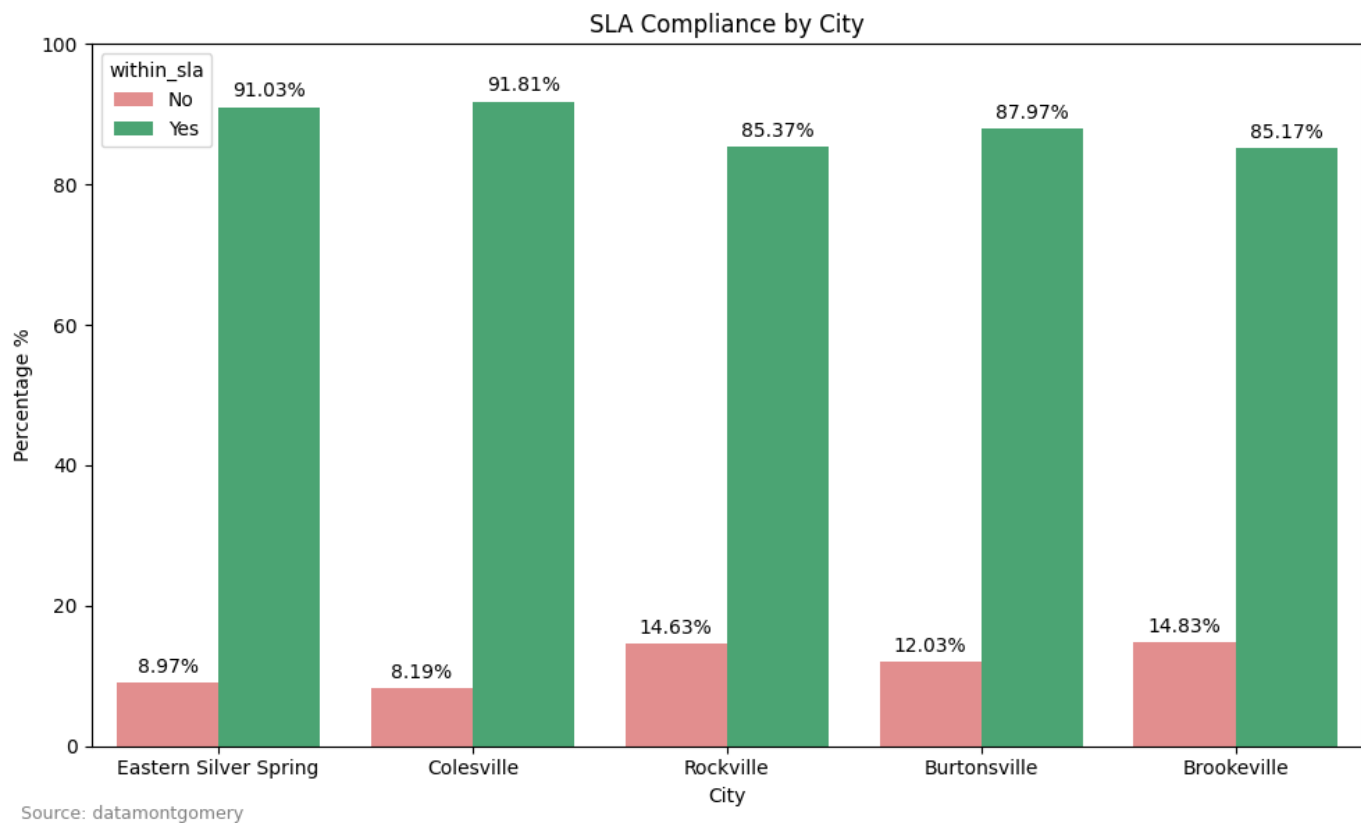
Source: datamontgomry, NOAA

Requests vs Temperature



Average Response Time by City

Chi-Squared test results show that there is a statistically significant association between city and SLA compliance ($p\text{-value} < 0.05$)



SLA Compliance by City

**Multiple regression
indicators:**

SLA Days
Population
Average Temperature
Income
Housing Units
Ethnicity
Median Age

R-Squared = 0.126

	precision	recall	f1-score	support
0	0.88	0.76	0.81	200
1	0.97	0.99	0.98	1829
accuracy			0.97	2029
macro avg	0.93	0.87	0.90	2029
weighted avg	0.96	0.97	0.96	2029

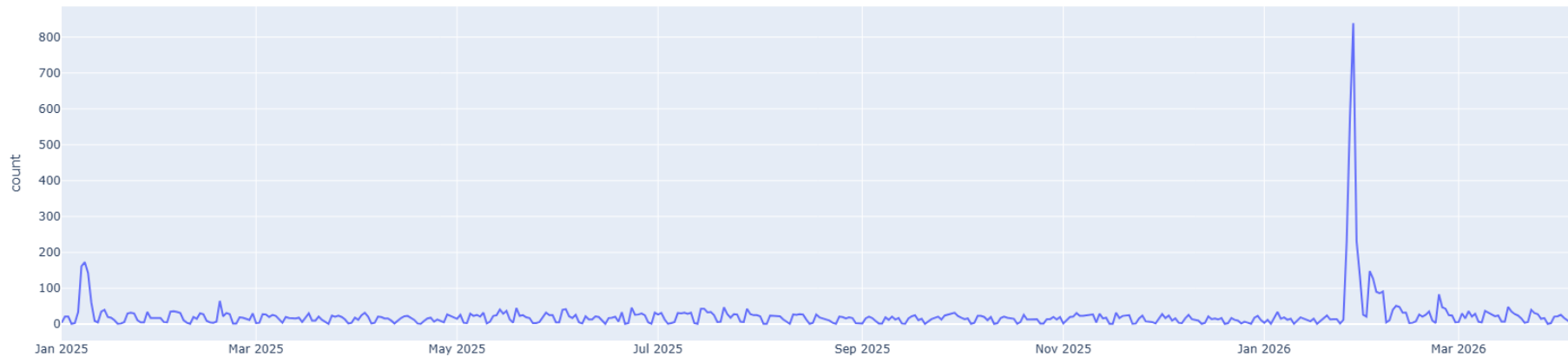
Classification Model for SLA Yes/No

97% accuracy - Dataset is imbalanced

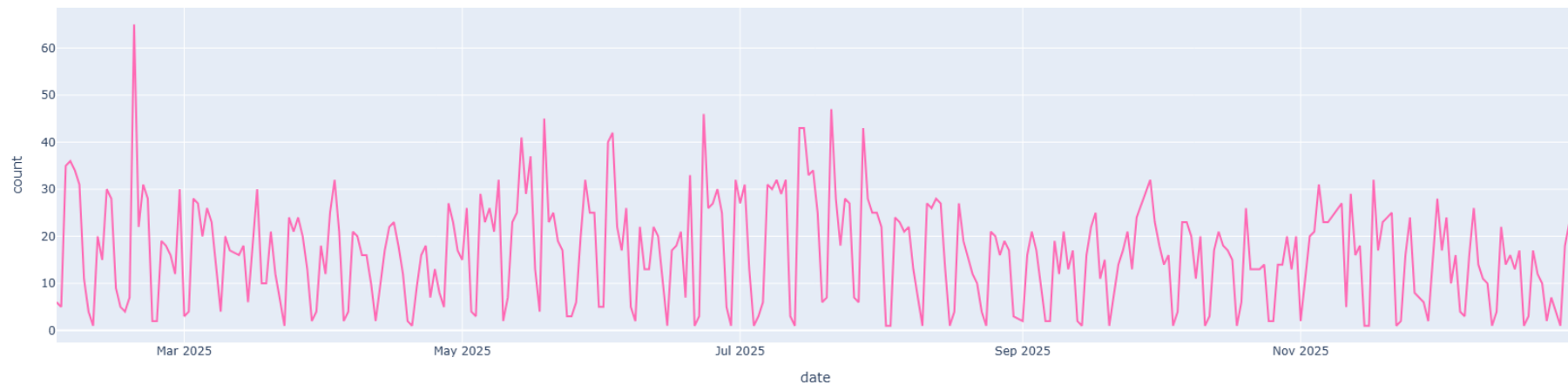
Better on for Yes (F1 = 0.98) compared to No (F1 = 0.81).

Effective at identifying Yes SLA cases, but with the recall of 0.76, it is less reliable in detecting No SLA cases.

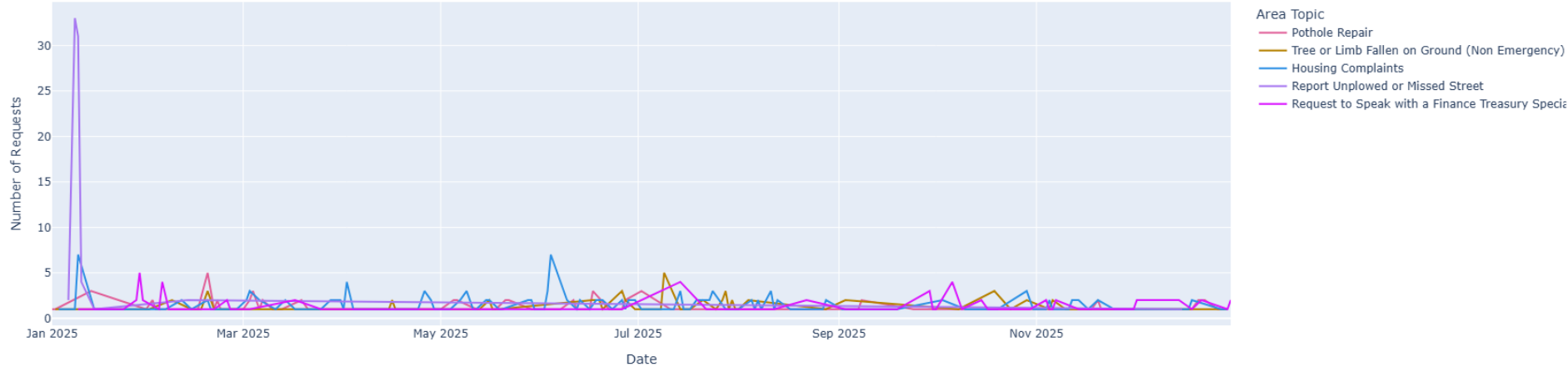
Requests Over Time



Requests Over Time (2025, Excluding January)



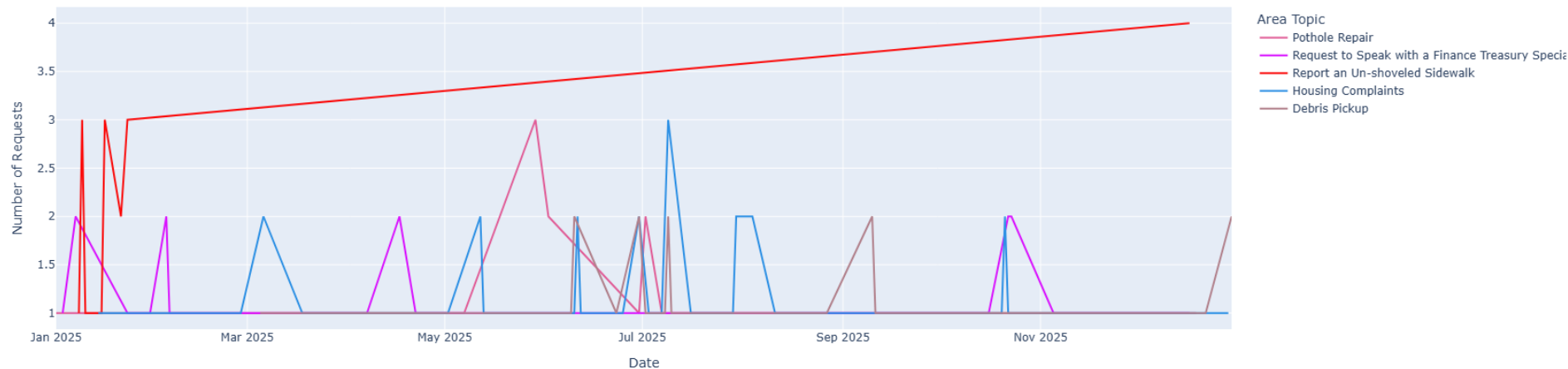
Requests Over Time in Rockville, Colored by Top Area Topics (2025)



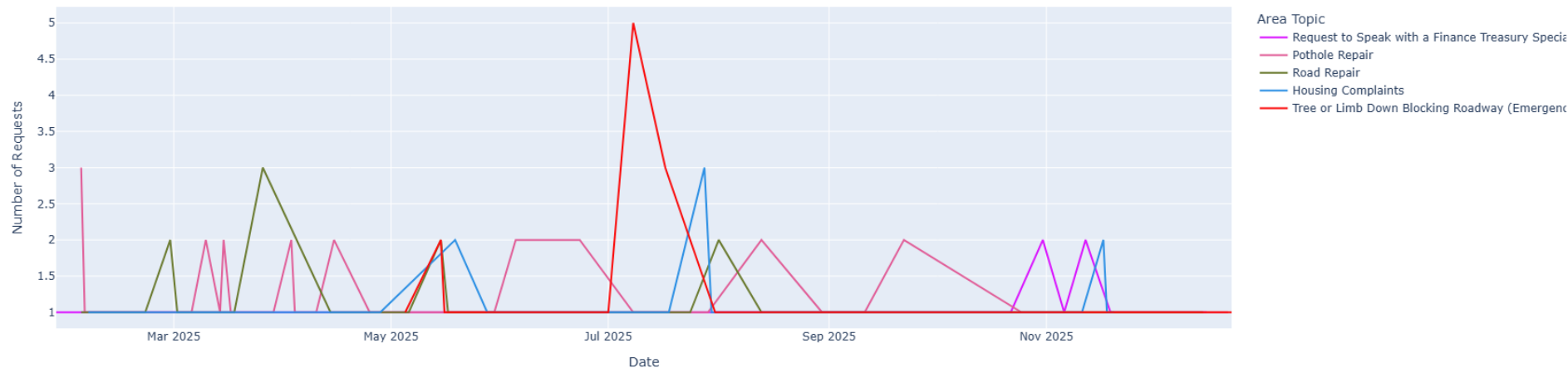
Source: datamontgomery, NOAA, maryland.gov, Data Commons

- Brookeville and Burtonsville has smaller population = less requests
- Peak of unplowed/missed street during January 2025 and peaks of housing complaints.
- Eastern Silver Spring had the highest peak for unplowed/missed street with a count of 75
 - Rockville had small peaks of street drainage and pothole repair requests
 - Colesville had small peaks of fallen trees/trees blocking the road

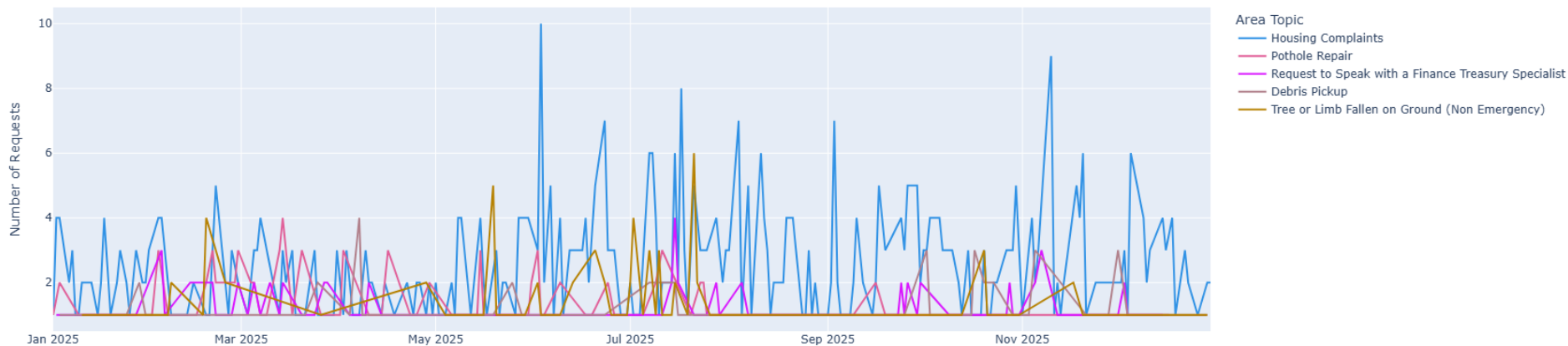
Requests Over Time in Burtonsville, Colored by Top Area Topics (2025)



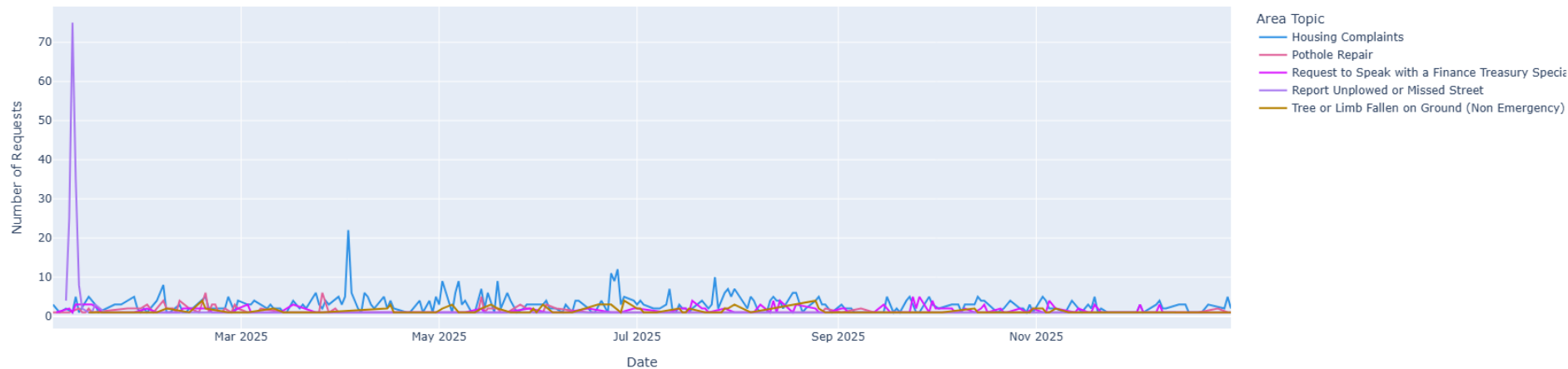
Requests Over Time in Brookeville, Colored by Top Area Topics (2025)



Requests Over Time in Colesville, Colored by Top Area Topics (2025)



Requests Over Time in Eastern Silver Spring, Colored by Top Area Topics (2025)



Takeaways



Weather impact is significant

- Service requests spike during colder temperatures and winter months
 - Snow-related issues

Geographic differences

- Rockville has the highest request rate per capita and quickest response time

Service patterns

- Highway Services dominate request types
- Request volume peak mid-week and drop on weekends

Model insights

- Regression model had low predictive power for variance in response time
 - SLA days only significant indicator
- Classification model had high accuracy (97%), but affected by class imbalance
 - Response time and SLA days are strongest predictors of SLA outcomes

Reflection

Challenges

Missing or incomplete data (mainly income datasets)

Class imbalance in classification model

Weak predictive power in regression model

What was rewarding

Successfully integrating datasets

Discovering meaningful trends

What I would change

Find a dataset about work procedures for

Use resampling techniques to address class imbalance

Use more advanced models for better prediction accuracy



Next Steps

Expand dataset to include:

- More ZIP codes and longer time range
- Additional variables

Recommendations!

Use findings to:

- Improve resource allocation during peak seasons
- Prepare for weather-related demand spikes
- Tailor to the unique needs and compliance patterns of each city



Github Account:
LNGHAN

Link:
<https://github.com/Inghan/MC311-Analysis>

Thank you to...



Professor Alraee

Professor Perine

Professor Park

Montgomery County Government

DataMontgomery team

DATA205 Classmates

Data Science Early College Cohort!!

